Sri Lanka Institute of Information Technology



IT3060 Human Computer Interaction

Assignment 1

User Research Techniques

Group No. 14
2025-WE-S2-14

Student ID	Name
IT22063564	RAJAWARDHANA M B G T N
IT22547774	ARIYARATHNA L M N R
IT22548078	CHATHURI M T K
IT22553478	DARMASENA M S T

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Member Details Workload Distribution.

Name	Student ID	Workload Distribution
RAJAWARDHANA M B G T N	IT22063564	Online payment page Dashboard SLBFE training program page SLBFE News and Alert page
ARIYARATHNA L M N R	IT22547774	Login page Foreign Job Vacancies pages SLBFE job bank profile page
CHATHURI M T K	IT22548078	Landing Page Complaint Submission Form Complaint Status View
DARMASENA M S T	IT22553478	Licensed foreign employment agencies (recruitment agencies) List of valid recruitment agencies as at (current date) List of temporarily suspended agencies/ cancelled agencies as at (current date)

Project Description.

The selected project for our study is the official website of the **Sri Lanka Bureau of Foreign Employment *(SLBFE)*,** which provides services and information for Sri Lankans seeking overseas employment. The site includes features such as worker registration, details about licensed agencies, complaint handling, and access to forms and guidelines. While it plays a vital role in supporting migrant workers, their families, and employers, the site also presents usability challenges that affect smooth navigation and accessibility for different users.

As part of our group activity, each member first selected a project individually and later formed a group of four to compare and finalize one project. After discussion, our team chose the SLBFE website, and the workload was divided so that each member is responsible for analyzing three interfaces. We then identified three main stakeholder groups migrant workers, foreign employment agencies, and SLBFE staff and created personas to represent their needs and experiences. In addition, we prepared contextual inquiry scripts to guide interviews with these stakeholders and recorded videos showing the issues they encountered when interacting with the website.

Finally, by analyzing these recordings, we identified at least ten usability problems and classified them into high, medium, and low severity levels. This process helps us better understand the website's strengths and weaknesses and provides clear directions for improving user experience. Through this structured approach, we aim to recommend design improvements that make the SLBFE website more user-friendly, accessible, and effective in serving its stakeholders.

Alternative Considered

Website Name	URL for Website
Top Jobs	https://topjobs.lk/
Youtube	https://www.youtube.com/
Fiverr	https://www.fiverr.com/
Upwork	https://www.upwork.com/

User Groups Identify

1. **Primary Users**: - These are the direct users of the system who frequently access and interact with it. They rely on website services for their personal needs.

   SLBFE Primary Users: -

   - Migrant Workers
   - Job Seekers

2. **Secondary Users**: - These users interact with the system occasionally, often for specific tasks, and sometimes with the help of others.

   SLBFE Secondary Users: -

- Licensed Recruitment Agencies

3. **Tertiary Users**: - These are users indirectly affected by the system. They may not use it daily but are influenced by its services and outcomes.

   SLBFE Tertiary Users: -

- SLBFE Staff (Marketing, Research, Welfare, Administration)

User Personas

Primary Users: Job Seeker (Sithmika Nanayakkara)



TECH STACK

- Languages
  C,Java,Kotlin,R,C#,Python,Java
- Database
  SQLServer,MySql,Mongodb,SQLite
- Web Development
  HTML,CSS,Bootstrap,Tailwind CSS

INTEREST

- Translated Books
- International Chemistry Books
- Information Technology Books

SKILLS

- Positive Leadership Qualities
- Communication Skills
- Teamwork
- Time Management
- Problem Solving

SITHMAKA NANAYAKKARA

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PROFILE

I am a passionate Software Engineering undergraduate with hands-on experience in both backend and frontend development. I've built secure RESTful APIs using Node.js, Express.js, and MongoDB with JWT authentication and microservices architecture.I'm eager to grow as a full-stack or backend developer and contribute to impactful, production-ready applications.

EDUCATION

- **BSc(Hons) in Information Technology Specializing in Software Engineering**
  2022-Present
- **GCE A/L Examination - Technology Stream**
  R / Rahula National School Godakawela
  1 B's 2 C's
- **English Certificate Course - Sabaragamuwa University**
  2023
- **Information Technology Diploma - ESoft Metro Campus**
  2023

PROJECTS

- **Cloud-Native Food Ordering & Delivery System using Microservices**
  Contributed to developing a full-stack food delivery platform with a focus on backend microservices for orders, restaurants, delivery, and payments.
- **Secure RESTful API for a Personal Finance Tracker System**
  Developed a secure backend API for managing transactions, budgets, financial goals, and reports with user roles, JWT authentication, multi-currency support, and notification handling.
- **Vehicle-Service-System**
  Built a full-stack vehicle parts store with backend services for product listings, cart operations, payment integration, and order management including tracking and status updates.

REFERENCES

Ms. K. G. M. Lakmali
Senior Lecturer
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Geethangali Wimalaratne
Senior Lecturer
SLIIT
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Secondary Users: Licensed Recruitment Agencies (Sarindu Samarasekara)



TECH STACK

- Languages
  Java,C++,C,Python,Kotlin,Spring boot
- Database
  SQLServer,MySql,Mongodb,Oracle
- Web Development
  HTML,CSS,JavaScript,Reactjs, Node js,Express.js

INTEREST

- Magazines
- English Books
- Information Technology Books

SKILLS

- Strong analytical skills
- Team Collaboration
- Positive Leadership Qualities

SARINDU SAMARASEKARA

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PROFILE

I am currently studying BSc(Hons) in Information Technology at the Sri Lanka Institute of Information Technology.Driven by a passion for innovation and learning,I aim to contribute effectively in fast-paced teams of technology experts while making both information technology and my IT career much more successful.

EDUCATION

- **BSc(Hons) in Information Technology Specializing in Data Science**
  2022-Present
- **GCE A/L Examination - Commerce Stream**
  ST. SYLVESTER'S COLLEGE
  Economics A
  Accounting A
  ICT C
- **ICBT CAMPUS**
  2022
  Diploma in English and ICT

PROJECTS

- **SkillHive**
  Web based skill learning and sharing platform developed as our third year group project.
- **LastMinutePantry**
  Near-Expired goods selling system as a solution for food waste problem in Sri Lanka as a personal project
- **Food order and Reservation System**
  Web based food order and reservation platform for perera and sons company as a personal project

REFERENCES

Prof. Prasad M. Jayaweera
Professor of Computer Science
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Tertiary Users: SLBFE Staff (Praveen Bakmeewewa)



TECH STACK

- Languages
  C++,Java,JavaScript
- Database
  SQLServer,Mongodb
- Web Development
  HTML,CSS,Tailwind CSS

INTEREST

- Novels
- Magazines
- English Books
- Information Technology Books

SKILLS

- Problem Solving
- Risk Management
- Decision Making
- Time Management
- Communication Skills

PRAVEEN BAKMEEWEWA

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PROFILE

I am a 23-year-old undergraduate (third-year) student at the Sri Lanka Institute of Information and Technology. I am highly motivated to work in a dynamic and competitive environment where I can continually enhance my skills and stay aligned with emerging technological trends in the industry.

EDUCATION

- **BSc(Hons) in Information Technology Specializing in Information Technology**
  2022-Present

PROJECTS

- **LASTMINUTESPANTRY**
  Near-Expired goods selling system as a solution for food waste problem in Sri Lanka as a group project.
  - Framework used - MERN Stack(React Native).
  - Languages used - JavaScript.
  - https://github.com/Tharu123122/ITPM_project.git
- **SkillHive**
  Web based skill learning and sharing platform developed as our third year group project.
  - Frameworks used - Spring Boot.
  - Languages used - Java, JavaScript.
  - Special Feature - Profile Assistant with voice command support.
  - https://github.com/hashini706050/skillsharing.git

REFERENCES

Geethangali Wimalaratne
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SLIIT
Email : geethanjali.w@sliit.lk

Usability Issues

1. IT22063564 RAJAWARDHANA M B G T N

Online payment page Dashboard	SLBFE training program page	SLBFE News and Alert page
• No clear navigation hierarchy • Overlapping purposes • Outdated UI design • Small button text and No language toggle • Bank promotion placement	• Complicated Navigation & Information Architecture • Inconsistent Design & Formatting • Lack of Mobile Responsiveness • No Search or Quick Access Options • Accessibility Shortcomings	• Too many menu items in the top navigation bar • Heavy color use (yellow, purple, grey, maroon) • Titles are long & inconsistent • No filtering or search option • Online Payment button at the top-right feels detached

2. IT22547774 ARIYARATHNA L M N R

Login page	Jobs Vacancies pages	SLBFE job bank profile page
• Different style from the main site • Minimal branding. No SLBFE logo, header, or footer • Asymmetrical layout • Poor alignment and Wasted space • Inconsistent button styles and Poor color contrast	• Unclear navigation structure • Duplicate/overlapping links • Spelling errors and Overuse of capital letters • Not mobile-friendly • Too much text emphasis and low readability	• Navigation Redundancy and Confusion • Non-Responsive and Outdated Design • Visual Clutter and Overcrowded Layout • Mismatch with other pages • Poor Visual Hierarchy and Unclear Field Instructions

3. IT22548078 CHATHURI M T K

Complaint Portal Entry Page	Complaint Submission Form	Complaint Actions/Status View
• Minimal Guidance / Lack of Context • Ambiguous Button Labels • Accessibility & Language Support • Lack of Visual Hierarchy / Button Differentiation • No Error Handling or Feedback Cues	• Overwhelming Form Complexity • Difficulty in Navigation / Scanning • Dropdowns Too Long & Not Searchable • No Input Validation or Real-Time Feedback • Language Overload / Bilingual Confusion	• Lack of Data Visibility by Default • No Search or Filter Mechanism • Unclear Column Context • Empty State Lacking Instructions • No Feedback on Data Loading or Errors

4. IT22553478 DARMASENA M S T

Search for Licensed Foreign Employment Agencies	Valid Recruitment Agencies List	Temporarily suspended recruitment
Outdated Visual Design Poor Label Clarity Inefficient Search Options No Mobile Responsiveness Unclear “Agency Directory A–Z” Navigation	Overcrowded Table Some fields have “” or “” without an explanation Multiple phone numbers in one cell without proper formatting No search within results users must scroll through 952 entries No Pagination	Overloaded text links No sorting/filtering option No explanation suspension / cancellation Table lack clear separation Status texting formatting inconsistent

Scripts for contextual inquiry.

Background description of the user	Questions to be asked from the user	Tasks to be given to the user
Job Seeker: Sithmika Nanayakkara Sithmika is a passionate Software Engineering undergraduate with hands-on experience in both backend and frontend development. I've built secure RESTful APIs using Node.js, Express.js, and MongoDB with JWT authentication and microservices architecture. I'm eager to grow as a full-stack or backend developer and contribute to impactful, production-ready applications.	1. Have you used the SLBFE website before? 2. How was your experience of finding job-related information? 3. Did you face any difficulties while filling out forms (e.g., complaint or registration)? 4. Did you try accessing the site on your phone? How was the experience? 5. Did you find the language options (Sinhala/Tamil/English) clear and easy to use?	• Ask the user to visit the SLBFE website and explain their first impression of the homepage • Ask the user to locate the "Job Opportunities" section on the SLBFE website and describe the process • Give the user a sample task: fill out a registration or complaint form on the website and note challenges such as unclear fields, dropdown usability, or validation issues. • Ask the user to open the website on their mobile device and perform a simple task (e.g., navigating menus, selecting a job post) and describe mobile usability issues. • Ask the user to switch between Sinhala, Tamil, and English versions of a specific page and report on translation accuracy, layout consistency, and ease of switching.

SLBFE Staff: Praveen Bakmeewewa Praveen is a 23-year-old undergraduate (third year) student at the Sri Lanka Institute of Information and Technology. I am highly motivated to work in a dynamic and competitive environment where I can continually enhance my skills and stay aligned with emerging technological trends in the industry.	1. Have you used the SLBFE website in your work before? 2. How easy is it for you to search for complaints submitted by workers or their families? 3. Does the system provide sufficient feedback after you update a complaint or take an action? 4. What challenges do you face while using the website daily? 5. What improvements would make the website more user-friendly for SLBFE officers?	• Ask the officer to log in to the SLBFE website and explain which parts they use most often in their daily work (e.g., complaints management, worker data). • Give the officer a task: search for a specific complaint using the available system features and describe the challenges they face (e.g., lack of filters, scrolling through results). • Ask the officer to update the status of a complaint or add notes and observe whether the system provides confirmation (e.g., success messages, status updates). • Ask the officer to perform a series of daily tasks (e.g., view complaints, process updates, navigate between modules) and note points of frustration or inefficiency. • Ask the officer to describe their “ideal” dashboard or workflow (e.g., advanced search, centralized dashboard, mobile-friendly design) and suggest features they believe should be added.

Licensed Recruitment Agencies: Sarindu Samarasekara Sarindu is a currently studying BSc (Hons) in Information Technology at the Sri Lanka Institute of Information Technology.Driven by a passion for innovation and learning,I aim to contribute effectively in fast-paced teams of technology experts while making both information technology and my IT career much more successful	1. Have you used the SLBFE website before for your agency's licensing or compliance needs? 2. How easy is it to find the information related to licensing and regulations? 3. Did you face any difficulties while submitting applications or updates through the website? 4. How was your experience using the website on different devices, such as mobile or tablet? 5. What improvements would make the website more useful for your agency?	• Ask the agency staff to log in and demonstrate how they typically access licensing or compliance-related resources. • Give the staff a task to locate a specific licensing guideline or regulation and describe the steps they take, highlighting navigation challenges. • Ask the staff to fill out (or simulate filling) a renewal application or update form and note issues such as unclear fields, long dropdowns, or timeouts. • Ask the staff to open the website on a mobile phone or tablet and attempt to complete a simple task (e.g., check license status) while noting difficulties. • Ask the staff to describe their "ideal agency dashboard" and the features they would expect (e.g., license tracking, searchable dropdowns, form validation, responsive design).

User Interview video Links.

i) **User 01:** Job Seeker Interview (Sithmika Nanayakkara)

ii) **User 02:** Licensed Recruitment Agencies (Sarindu Samarasekara)

iii) **User 03:** SLBFE Staff (Praveen Bakmeewewa)

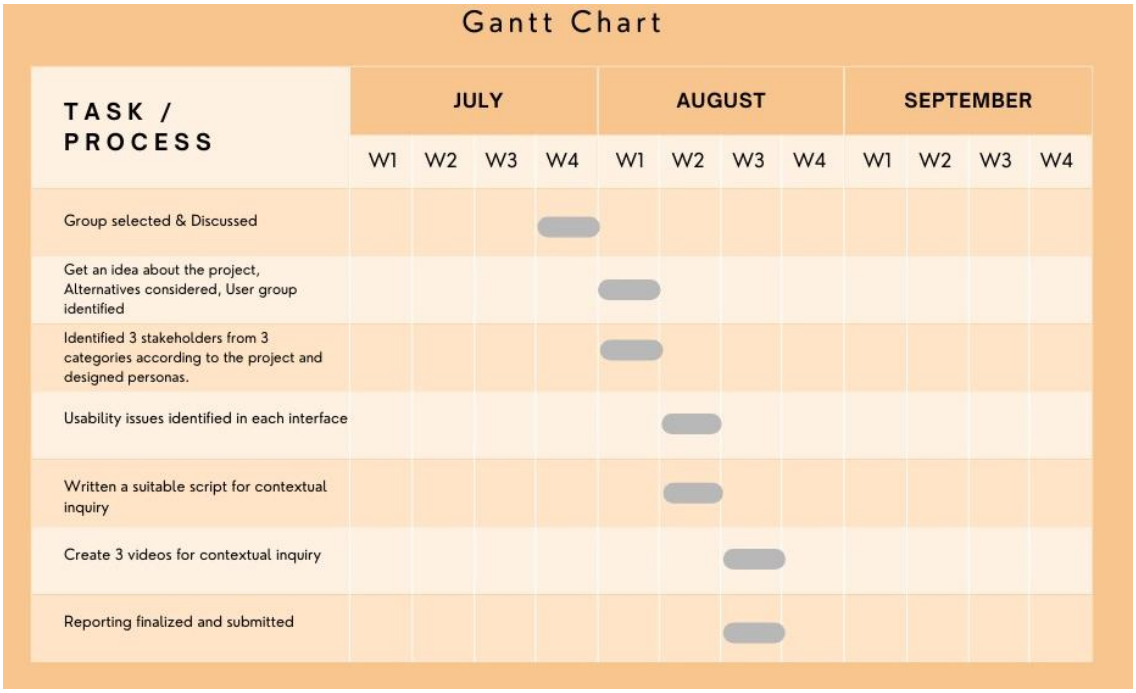
Usability issues identified based on contextual inquiry.

Users	Useability Issues
Job Seeker	• Bad user interfaces • Bad color combinations • Nav bar is not in correct order. • Give yourself additional time to access the website. • Upgrade to correct order and proper user interfaces
Licensed Recruitment Agencies	• More white spaces and not consistency • Unclear “Agency Directory A–Z” Navigation • No search within results users must scroll through 952 entries • No sorting/filtering option • Overloaded text links and tables
SLBFE Staff	• Users cannot understand the purpose of this site at once • Poor Navigations • No Search or Filter Mechanism • Overwhelming Form Complexity • Not organized well

Ratings of the Usability Problems as High Medium and Low.

Usability Issue	Rating
There’s no well-arranged form for users to enter their details in Login and Registration page	**High**
No Registration option and difficult to find the Register	**High**
Login and Register functions cannot be on the same page	**High**
It is difficult to find out where the user's information is changed	**Medium**
Registration and logging forms are not clear	**High**
Minimal Guidance / Lack of Context	**Low**
No Mobile Responsiveness	**High**
No Search or Filter Mechanism	**High**
Overwhelming Form Complexity	**Medium**
No Pagination	**Medium**
Poor Visual Hierarchy and Unclear Field Instructions	**High**

Grant Chart


Gantt Chart
TASK / PROCESS
JULY
AUGUST
SEPTEMBER
W1 W2 W3 W4 W1 W2 W3 W4 W1 W2 W3 W4
Group selected & Discussed
Get an idea about the project, Alternatives considered, User group identified
Identified 3 stakeholders from 3 categories according to the project and designed personas.
Usability issues identified in each interface
Written a suitable script for contextual inquiry
Create 3 videos for contextual inquiry
Reporting finalized and submitted

References

https://www.interaction-design.org/literature/article/accessibility-usability-for-all

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https://thomasdigital.com/bad-websites